

Problems from 11.1

The *fundamental period* is the smallest positive period. Find it for

1. $\cos x, \sin x, \cos 2x, \sin 2x, \cos \pi x, \sin \pi x, \cos 2\pi x, \sin 2\pi x$
2. $\cos nx, \sin nx, \cos \frac{2\pi x}{k}, \sin \frac{2\pi x}{k}, \cos \frac{2\pi nx}{k}, \sin \frac{2\pi nx}{k}$
3. If $f(x)$ and $g(x)$ have period p , show that $h(x) = af(x) + bg(x)$ (a, b , constant) has the period p . Thus all functions of period p form a **vector space**.
4. **Change of scale.** If $f(x)$ has period p , show that $f(ax)$, $a \neq 0$, and $f(x/b)$, $b \neq 0$, are periodic functions of x of periods p/a and bp , respectively. Give examples.

Sketch or graph $f(x)$ which for $-\pi < x < \pi$ is given as follows.

6. $f(x) = |x|$
7. $f(x) = |\sin x|, f(x) = \sin |x|$
8. $f(x) = e^{-|x|}, f(x) = |e^{-x}|$
9. $f(x) = \begin{cases} x & \text{if } -\pi < x < 0 \\ \pi - x & \text{if } 0 < x < \pi \end{cases}$
10. $f(x) = \begin{cases} -\cos^2 x & \text{if } -\pi < x < 0 \\ \cos^2 x & \text{if } 0 < x < \pi \end{cases}$

Problems from 11.2

Are the following functions even or odd or neither even nor odd?

3. Sums and products of even functions
4. Sums and products of odd functions
5. Absolute values of odd functions
6. Product of an odd times an even function
7. Find all functions that are both even and odd.

EXAMPLE 5 Sawtooth Wave

Find the Fourier series of the function (Fig. 268)

$$f(x) = x + \pi \quad \text{if } -\pi < x < \pi \quad \text{and} \quad f(x + 2\pi) = f(x).$$

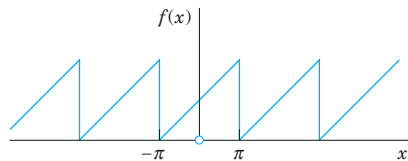


Fig. 268. The function $f(x)$. Sawtooth wave

Find the Fourier series of the function below. Avoid calculating integrals, reuse as much of the sawtooth result as possible.

10.

